# CONNIE XU

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**Education**

**Harvard University**
Ph.D. Health Policy and Economics, 2021 to 2027 (expected)
**University of Chicago**
B.A. in Economics (with Honors), 2019
B.A. in Statistics, 2019

**Fields**

Innovation Economics
Health Economics
Labor Economics

**References**

| **Amitabh Chandra (chair)** | **Luca Maini** | **Kyle Myers** |
|---|---|---|
| achandra@hbs.edu | maini@hcp.med.harvard.edu | kmyers@hbs.edu |

**Fellowships & Awards**

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| NBER Fiscal and Economic Effects of Innovation and Productivity Policies Fellow | 2026-2027 |
| National Science Foundation Graduate Research Fellowship Program | 2024-2027 |
| NBER Pre-doctoral Fellowship in Aging & Health | 2023-2024 |
| Harvard Kennedy School Distinction in Student Teaching Award | 2024 |
| Agency for Healthcare Research and Quality T32 Trainee | 2021-2023 |
| Phi Beta Kappa | 2019 |
| UChicago Careers in Health Professions Health Policy Scholar | 2016-2019 |

**Teaching**

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| Instructor, *Essentials for the Profession Course (Nursing Homes)*, Harvard Medical School | 2025 |
| Teaching Fellow, *Empirical Methods II (Econometrics)*, Harvard Kennedy School | 2024 |

**Research Assistantships**

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| Graduate Research Assistant to Professor Amitabh Chandra, *Harvard Business School* | 2022-2024 |
| Graduate Research Assistant to Professor Timothy Layton, *Harvard Medical School* | 2022-2024 |
| Pre-doctoral Research Fellow to Professor Amy Finkelstein, *MIT* | 2019-2021 |
| Microeconomics Research Intern, *Federal Reserve Bank of Chicago* | 2018 |

**Job Market Paper**

**Managing the Rising Cost of Science: Supplier Consolidation in the Academic Life Sciences**, *with Ruby Zhang*


Scientists cannot easily substitute away from the inputs their experiments depend on or pass on higher costs, leaving them vulnerable to rising prices in increasingly concentrated markets for life sciences inputs. How much science is lost when these inputs become more expensive, and which scientists are better able to adjust? We use the 2014 merger of two large life sciences suppliers as a shock to the prices principal investigators (PIs) pay for these inputs, drawing on novel purchase-level procurement records from U.S. public universities linked to PIs' publications. Prices in affected markets rose by almost 20 percent, while quantities declined only slightly and temporarily, consistent with highly inelastic demand. PIs facing larger increases in their input costs published less. For the average PI, input costs rose by about 2.8 percent and research output fell by about 1.2 percent, implying an elasticity of −0.42. Higher input prices therefore reduce scientific production, not just raise research costs. The output losses fell disproportionately on early-career PIs. Financial resources do not fully explain this difference. Instead, our findings suggest that experience helps PIs adjust to higher prices through tacit knowledge accumulated during training and years of running a lab.

**Working Papers**

**Person and Place Effects in Scientific Discovery**, *with Amitabh Chandra*

How much of a scientist's research output is attributable to individual talent, and how much to the institution where they work? Because markets systematically under-provide fundamental science, the answer matters for how societies allocate the resources that substitute for missing market incentives, how the scientific community evaluates researchers for promotion and grants, and whether the current distribution of scientists across institutions maximizes knowledge production. We study this question using citation-weighted publications in leading life-science journals and a movers design that exploits variation in the output of nearly 40,000 scientists who change institutions. We find that 50 to 60 percent of the variation in a scientist's research output is attributable to institutional factors. After correcting for limited mobility bias, we document positive assortative matching between scientists and institutions (a correlation of 0.3 between person and place effects), a necessary condition for maximizing aggregate knowledge production under the multiplicative structure of our model. Around two-thirds of the institutional effect is explained by the presence of star researchers. A scientist's output responds to both the total output of their institution and its funding level, but not to the size of the local research cluster, suggesting that institutional effects operate through the human capital environment within an institution rather than through agglomeration or research funding alone. The symmetry of effects for upward and downward movers implies that institutional advantages do not permanently transfer to scientists who leave. These effects have not diminished despite technologies facilitating remote collaboration, are robust to detailed field-by-year controls, and raise the possibility that similar patterns extend to other areas of fundamental science.

**Returns to Science of Centrally Provisioned GPUs and Optimal Allocation**, *with Ruby Zhang*

Should federal infrastructure grants go to universities with higher productivity, or to universities with tighter budgets that cannot self-finance capital investments? We study the allocation of federal computing grants to U.S. universities by estimating the returns to science of centrally provisioned GPUs and modeling the crowd-out of private investment. We construct a novel panel of CPU and GPU capacity at 53 U.S. R1 universities with publication counts as our output measure. Exploiting large and lumpy increases in GPU capacity, we find that AI and deep-learning publications respond more at less-resourced institutions (restricting to incumbent researchers). To decompose the effect of capital versus labor, we structurally estimate a production function where multiple fields share a central cluster, and then derive universities' investment responses to external grants. Holding labor fixed, the observed National Science Foundation (NSF) portfolio captures 39% of the publication gain attainable with the same budget. The publication-maximizing allocation shifts dollars from universities with medium-sized endowments toward both tails. Since the NSF has mandates beyond maximizing publications, we find that the NSF's observed allocation is more geographically dispersed and consistent with maximizing proportional rather than absolute gains across universities.

**The Ripple Effects of Health Care Entry Regulation**, *with Yunan Ji and Parker Rogers*

We examine how regulatory entry costs for healthcare inputs shape market structure, pricing, and bargaining dynamics throughout the healthcare supply chain. Using a difference-in-differences approach exploiting the FDA's 2016 removal of pre-market clearance requirements for certain medical devices, we find increased manufacturer entry and lower negotiated device prices between manufacturers and hospitals. These reduced input prices propagate downstream, modestly decreasing procedure prices negotiated with insurers and ultimately lowering patient out-of-pocket costs. We then develop and will estimate a structural model that jointly captures manufacturer entry decisions and simultaneous bargaining between hospitals, manufacturers, and insurers. Our proposed counterfactual simulations will quantify how tightening entry regulations or increasing horizontal consolidation would ripple through the supply chain.

**Invited Presentations**

NBER Productivity Seminar 2026
European Association for Research in Industrial Economics Conference

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| | American Society of Health Economists Conference | |
| | Wharton Innovation Doctoral Symposium | |
| | NBER Assessing the U.S. Medical Innovation System | |
| | Harvard Seminar in the Economics of Science & Engineering | 2025 |
| | NBER Place-Based Policies and Entrepreneurship | 2024 |
| **Academic Service** | **Referee:** *The Journal of the American Medical Association* | |
| | **Organizer:** Harvard Health Care Policy Health Economics Seminar | |
| | **Peer Mentor:** Harvard Health Policy PhD Program | |
| **Research Grants** | The Lab for Economic Applications and Policy Award, *with Ruby Zhang* | 2026-2027 |
| | John and Kiendl Gordon Economics Support Fund, *with Ruby Zhang* | 2025-2026 |
| | The Lab for Economic Applications and Policy Award, *with Ruby Zhang* | 2024-2025 |
| | NBER Center for Aging and Health Research Pilot Grant, *with Amitabh Chandra and Timothy Layton* | 2023-2024 |
| **Languages** | English (native), Mandarin (fluent) | |
| **Citizenship** | United States | |